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Global Health EDCTP3
and its members

STUDY DESIGNS

STEAM-3D
Sustainable Training in Epidemiological Analysis and Modelling for Data-Driven Decision Making in Africa
SOUTH AFRICA | MALAWI | MOZAMBIQUE | CAMEROON | BELGIUM

Learning Outcomes

- Different types of study designs

What is a study design?

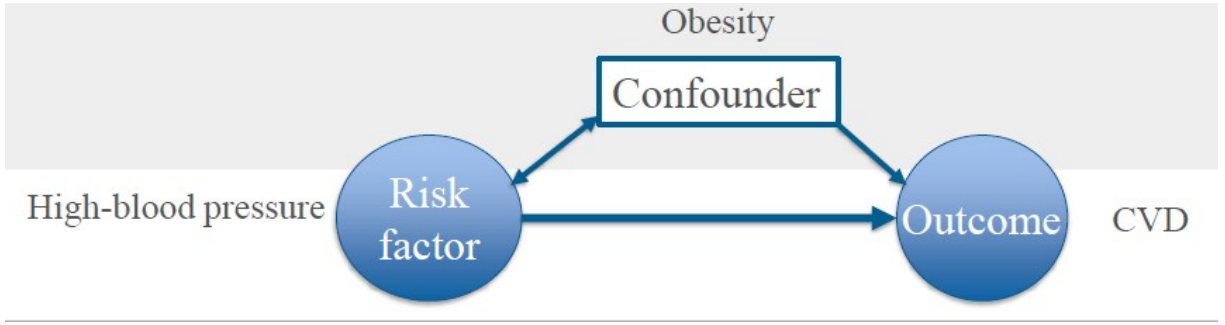
- A specific plan or protocol for conducting the study, which allows the investigator to translate the conceptual hypothesis into an operational one.
- An appropriate study design is essential in ensuring validity of the results that you eventually report
- Research question needs to be clearly defined
  - Literature review
  - Narrow down research question to something specific that can be reasonably studied

Factors to consider

- Confounding
- Bias
- Length of time to conduct study
- Expense
- Broad vs specific question

Confounding variables

- Confounding occurs when association due to three factors: exposure, outcome of interest and third factor associated with outcome and exposure, but not causal factor itself .
- Can lead to erroneous conclusions about the relationship between the independent and dependent variables.
- Deal with confounding variables by controlling for them: matching, randomizing, or statistical control.



Bias

- In research, bias occurs when the way a study is designed or carried out causes an error in the results and conclusions
- Bias can occur at any phase of research, including study design or data collection, as well as in the process of data analysis and publication
- Can cause estimates of association to be smaller or larger than true association

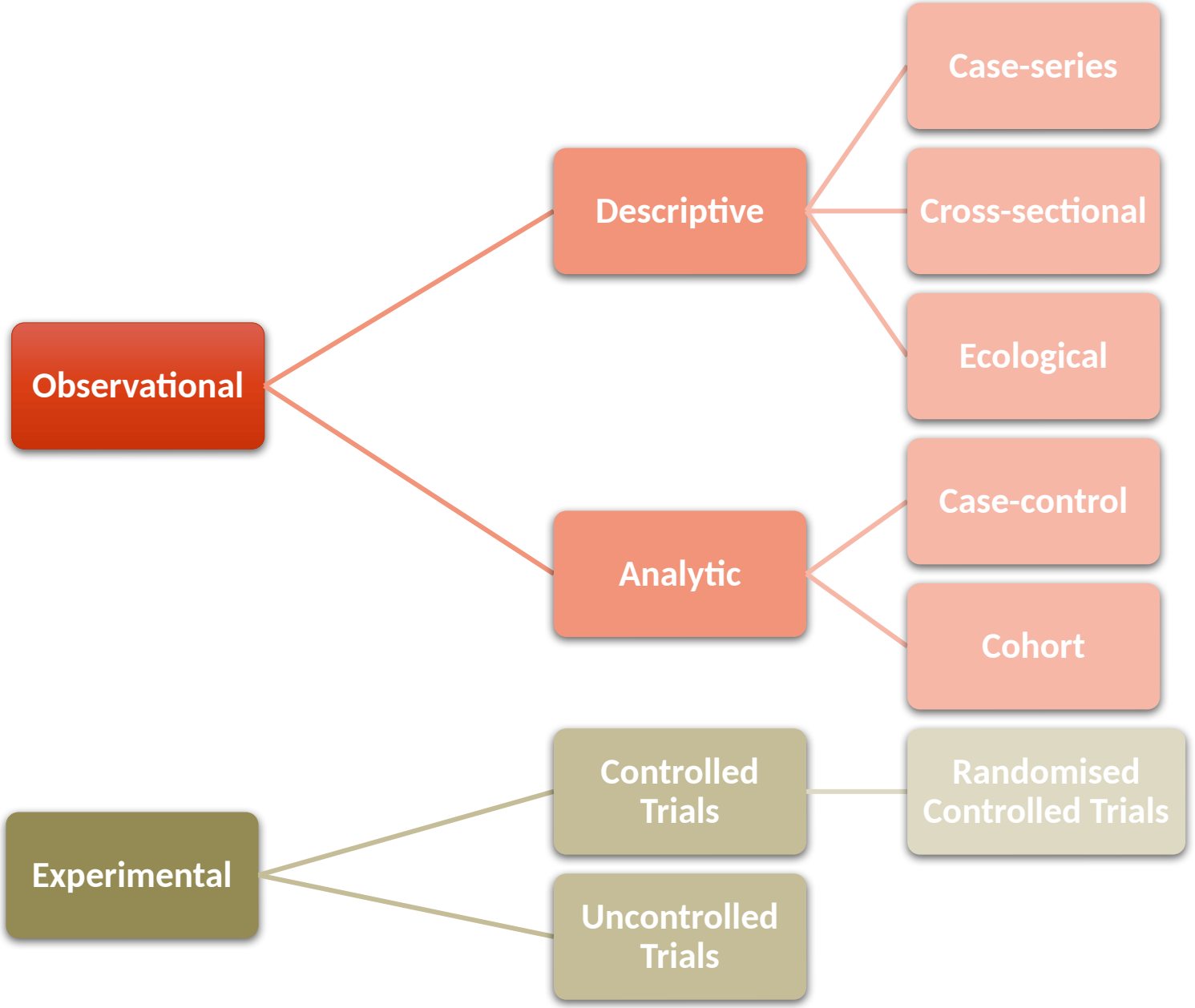
➢Consider the degree to which bias was prevented

- proper study design and implementation

Types of Studies

- Observational: researcher observes and systematically collects information but does not change the subjects (animals, humans, reagents) being observed i.e. no interventions

- Experimental studies: sample of subjects with a condition and who meet other selection criteria are randomly allocated to receive either the experimental treatment or control treatment (standard treatment or placebo)- some sort of intervention takes place


Observational
Descriptive
Case-series
Cross-sectional
Ecological
Analytic
Case-control
Cohort
Experimental
Controlled Trials
Randomised Controlled Trials
Uncontrolled Trials

Observational Studies

- Descriptive: describes data and characteristics about the population or phenomena being studied e.g. estimating the **prevalence of HIV** in pregnant women attending antenatal clinics.

- Analytic: used to test hypotheses e.g. is the risk of developing tuberculosis higher among people with diabetes compared with people without diabetes?

- Used when:
  - Want to collect descriptive information
  - Want to report on causes of a problem without disturbing natural setting
  - When experiments are not possible
  - When it is unethical/unacceptable to do an experiment

Types of observational studies

- Case-series studies
- Case-control studies
- Cohort studies
- Cross-sectional studies
- Ecological studies

Case-series studies

- Simple descriptive account of interesting characteristics observed in a group of subjects

- Generally involve subjects seen over a relatively short period of time

- Useful for:
  - Studying predictive symptoms, signs and tests
  - Creating case definitions
  - Clinical education, audit and research
  - Establishing safety profiles

Case-series: advantages & disadvantages

✓ Easy to write

✓ Observations useful for designing a study to evaluate causes or explanations

x Susceptible to many biases

➢ Generally: hypothesis-generating and not conclusive

Case-series: Example

Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China

Chaolin Huang, Yeming Wang*, Xingwang Li*, Lili Ren*, Jianping Zhao*, Yi Hu*, Li Zhang, Guohui Fan, Jiuyang Xu, Xiaoying Gu, Zhenshun Cheng, Ting Yu, Jiaan Xia, Yuan Wei, Wenjuan Wu, Xuelei Xie, Wen Yin, Hui Li, Min Liu, Yan Xiao, Hong Gao, Li Guo, Jungang Xie, Guangfa Wang, Rongmeng Jiang, Zhancheng Gao, Qi Jin, Jianwei Wang†, Bin Cao†*

Summary

Background A recent cluster of pneumonia cases in Wuhan, China, was caused by a novel betacoronavirus, the 2019 novel coronavirus (2019-nCoV). We report the epidemiological, clinical, laboratory, and radiological characteristics and treatment and clinical outcomes of these patients.

Methods All patients with suspected 2019-nCoV were admitted to a designated hospital in Wuhan. We prospectively collected and analysed data on patients with laboratory-confirmed 2019-nCoV infection by real-time RT-PCR and next-generation sequencing. Data were obtained with standardised data collection forms shared by WHO and the International Severe Acute Respiratory and Emerging Infection Consortium from electronic medical records. Researchers also directly communicated with patients or their families to ascertain epidemiological and symptom data. Outcomes were also compared between patients who had been admitted to the intensive care unit (ICU) and those who had not.

Findings By Jan 2, 2020, 41 admitted hospital patients had been identified as having laboratory-confirmed 2019-nCoV infection. Most of the infected patients were men (30 [73%] of 41); less than half had underlying diseases (13 [32%]), including diabetes (eight [20%]), hypertension (six [15%]), and cardiovascular disease (six [15%]). Median age was 49·0 years (IQR 41·0–58·0). 27 (66%) of 41 patients had been exposed to Huanan seafood market. One family cluster was found. Common symptoms at onset of illness were fever (40 [98%] of 41 patients), cough (31 [76%]), and myalgia or fatigue (18 [44%]); less common symptoms were sputum production (11 [28%] of 39), headache (three [8%] of 38),

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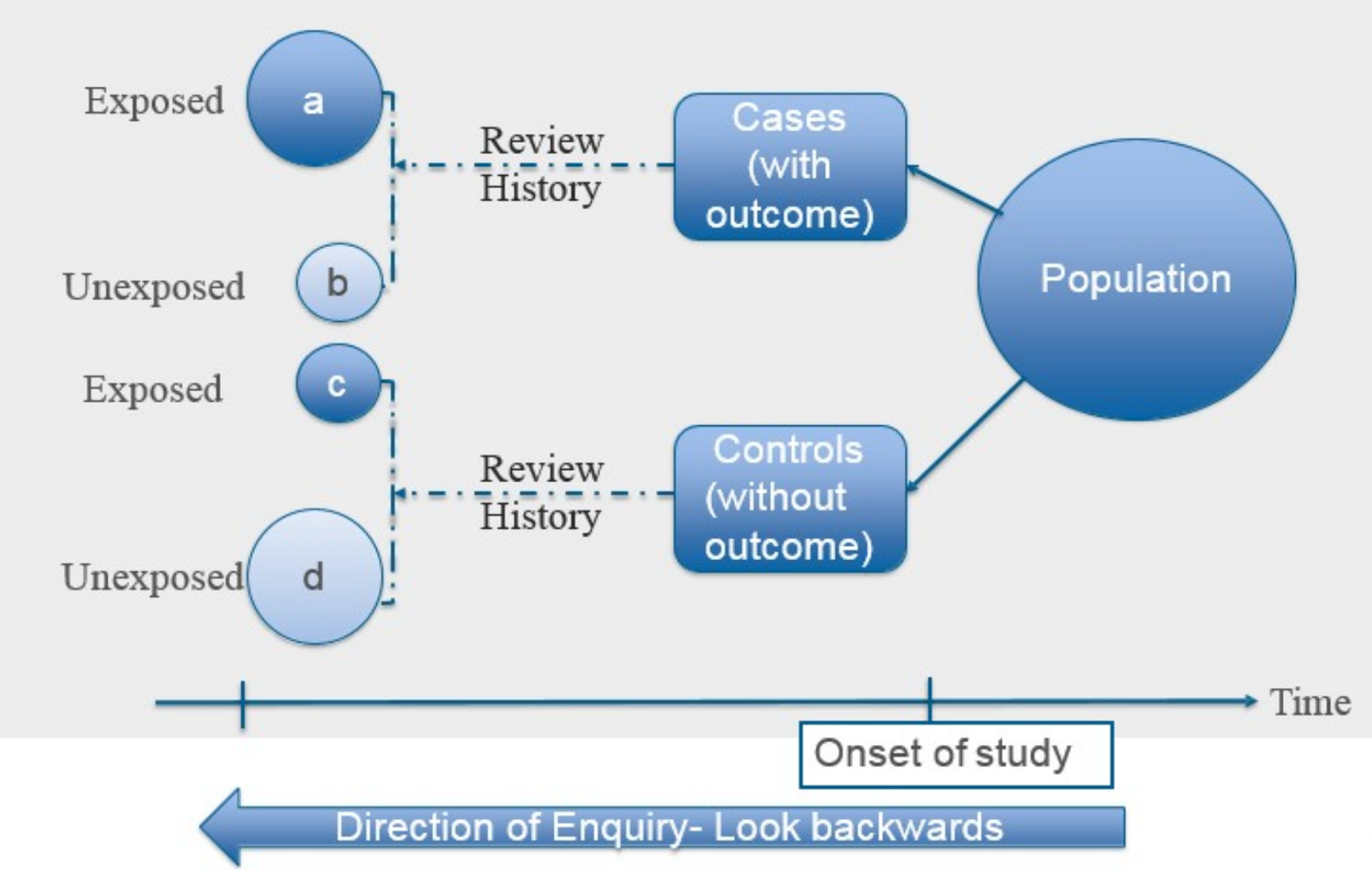
See Comment pages 469 and 470

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Case-control studies



- Begin with absence or presence of outcome, then look backward in time to try detect possible causes or risk factors- retrospective study

- Enquiry covers a period of time

- Things to consider:
  - Confounders
  - Unsure which came first- disease or exposure?
  - Cannot calculate prevalence or incidence
  - Odds ratios can be calculated

Case-control: advantages & disadvantages

- ✓ Appropriate for studying rare diseases or events
- ✓ Examining conditions that develop over long period
- ✓ Quick and least expensive to undertake
- ✓ For investigators who need data prior to writing proposal for more time-consuming study
- x Large number of possible biases and errors
- x Depend on high-quality existing records
- x Compromise between what researcher wishes to study and what they are able to study
- x Selection of appropriate control group

Case-control: Example

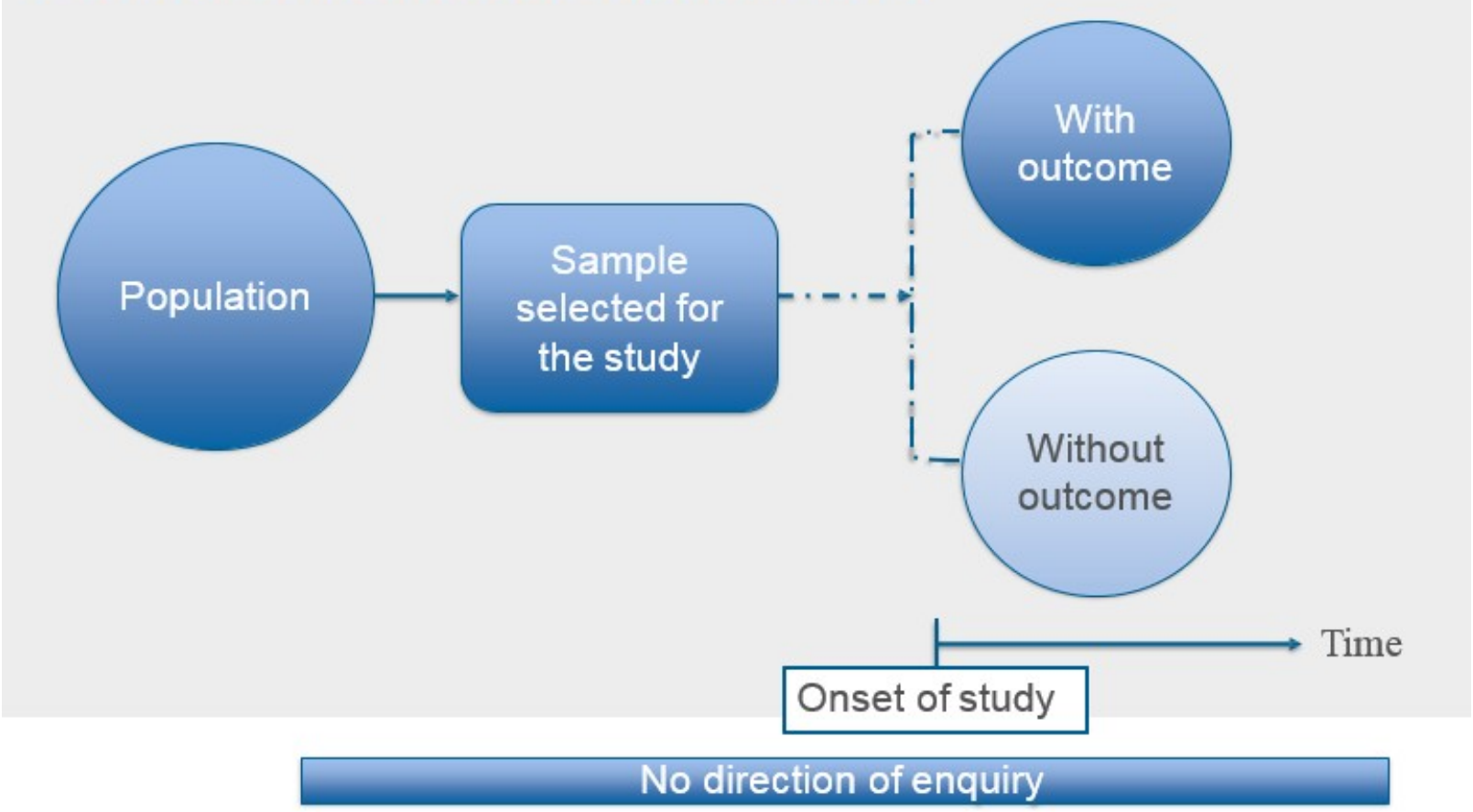
Case-Control Study Example (NEJM)

A Case-Control Study of HIV Seroconversion in Health Care Workers after Percutaneous Exposure (1997)

- Cases: Healthcare workers who acquired HIV after needlestick exposure.
- Controls: Exposed healthcare workers who remained HIV-negative.
- Compared characteristics of the exposure.
- Identified major risk factors for occupational HIV transmission.
- Informed global recommendations for HIV post-exposure prophylaxis.

Key lesson:
Starting with people who developed infection (cases) and those who did not (controls), researchers looked backwards to identify exposure characteristics associated with transmission.

Cross-sectional studies



- Also known as surveys, epidemiological studies, prevalence studies
- Analyze data collected on a group of subjects at one point of time
  - Disease and exposure status measured simultaneously
- Typically used for descriptive studies (associations)
- Beware: causation and correlation

Causation vs correlation

- Correlation does not imply causation
- Causation: an action or occurrence causes another e.g. smoking causes lung cancer
- Correlation: smoking is correlated with cardiovascular disease

Therefore; showing a variable is correlated with another does not mean it causes it

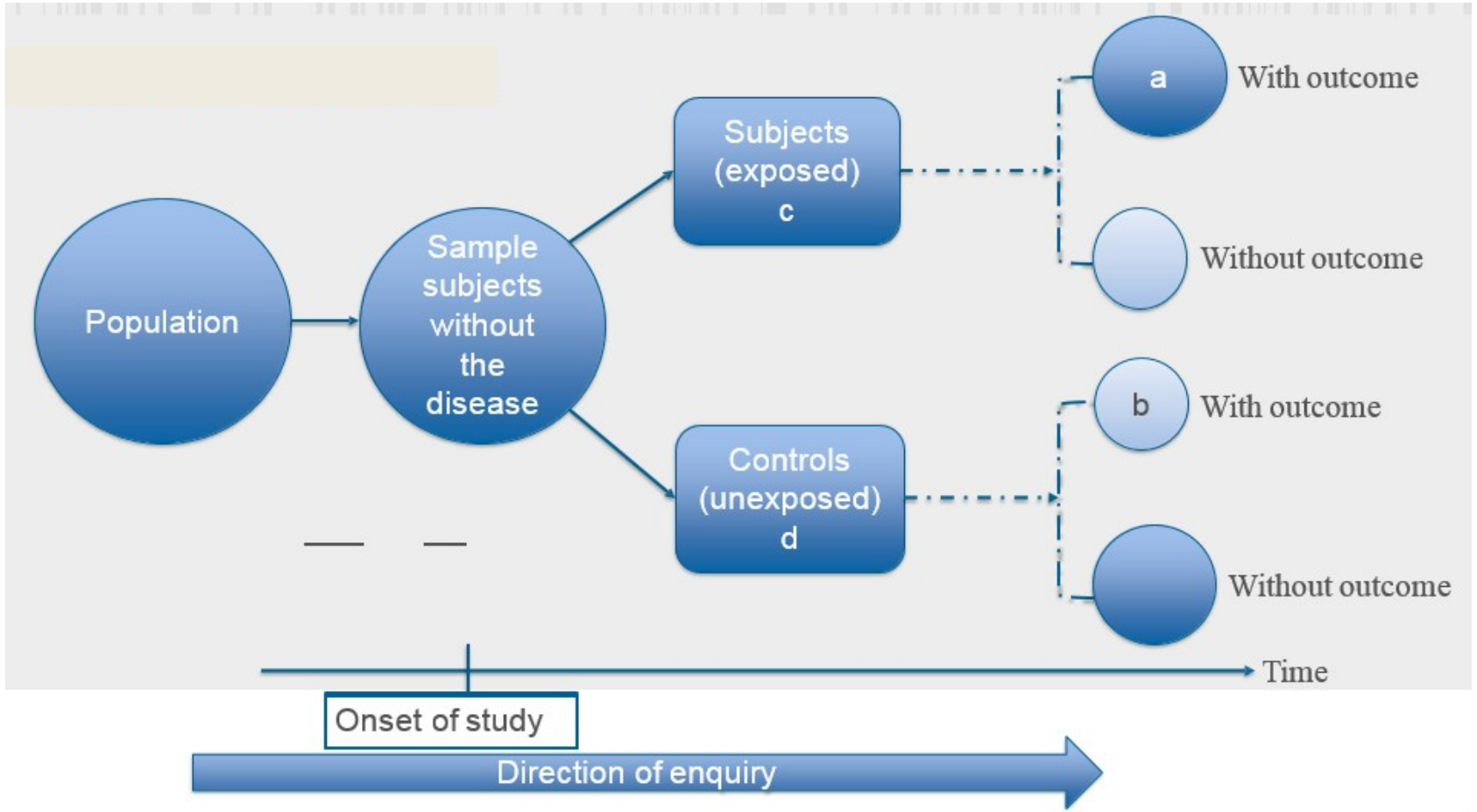
Cross-sectional studies: advantages & disadvantages

✓ Best for determining the status quo of a disease/condition e.g. prevalence of HIV

✓ Quick and relatively inexpensive

x Provide “snapshot in time” not whole disease process

Cohort studies



- Cohort- group of subjects who share common characteristics or experiences within a defined period
- Comparison group may be drawn from general population or be another (unexposed) cohort of persons
- Allows you to study changes and establish time-sequence in which things occur
- Incidence rates can be calculated and/or risk ratios

Cohort studies: advantages & disadvantages

✓For studying causes of a condition, course of a disease or risk factors

✓Strong evidence for possible causes and effects

✓Investigators can control for many sources of bias

x Costly e.g. for diseases that develop over long periods

x Cannot determine causation

x Loss to follow-up in ‘people’ studies

Ecological studies

- Studies of risk-modifying factors on health or other outcomes based on populations defined either geographically or temporally
  - Collect data in the aggregate rather than individual level i.e. level of a neighborhood, city, state or country
- Used when information at individual level not available or focus is on community well-being e.g. herd health status/immunity
- Cohen et al (2003) looked at boarded-up housing on rates of gonorrhea in 107 cities- results consistent with hypothesis that physical deterioration of neighborhoods leads to social isolation and unsafe health practices

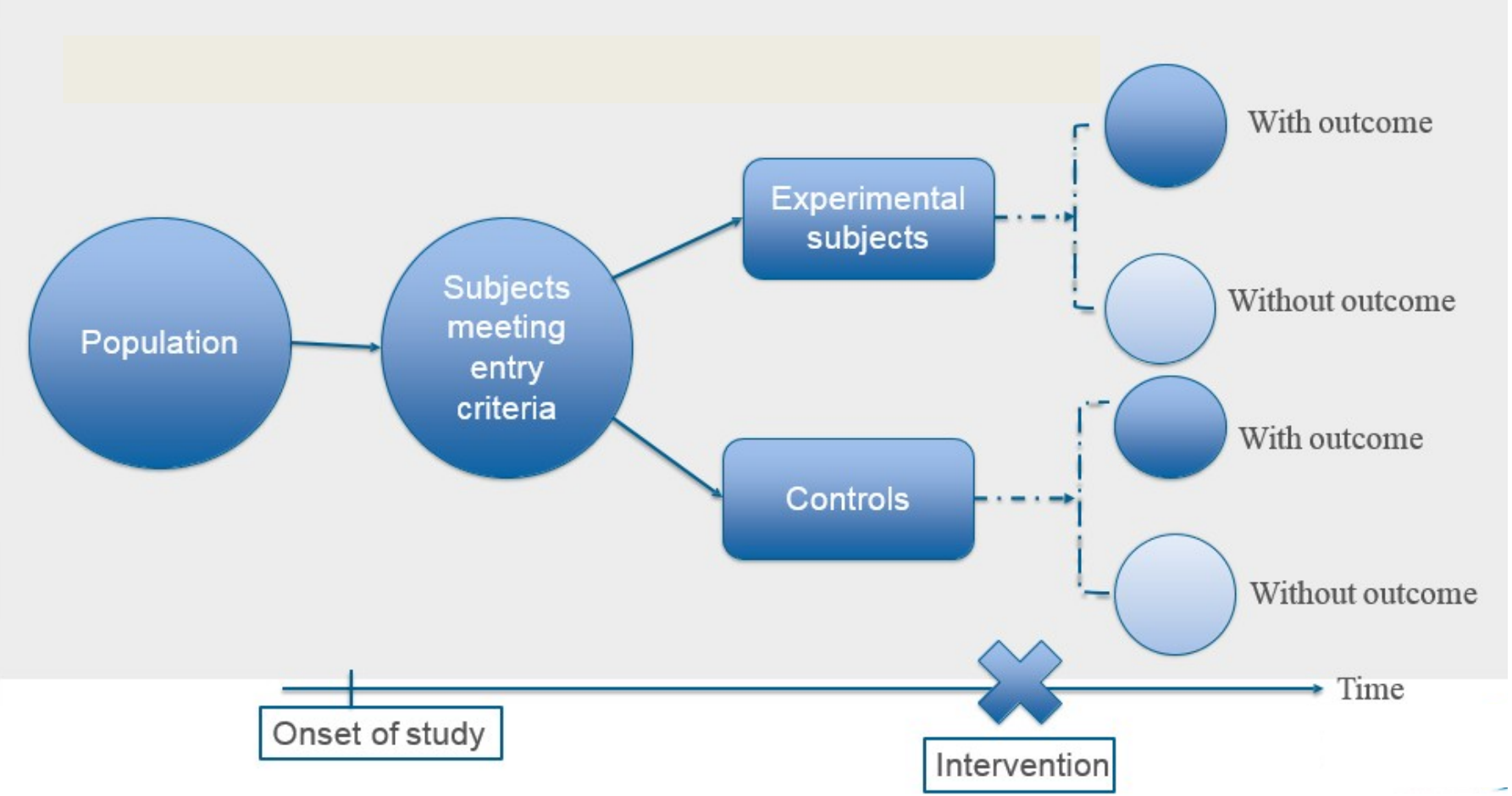
➢ However: data was not collected from individuals- could be other confounding factors- ecological fallacy

Experimental Studies

- Controlled trials: studies in which experimental drug or procedure is compared with another drug or procedure

- Uncontrolled trials: studies in which investigators experience with drug or procedure is described but not formally compared with another treatment

Randomised controlled trials (RCTs)



- Sample of subjects with condition, who meet other selection criteria randomly allocated to receive either treatment or control treatment (placebo or standard treatment)
- Random allocation reduces bias and confounding
- Blinding
- Absolute risk reduction/relative risk reduction

RCTs: advantages & disadvantages

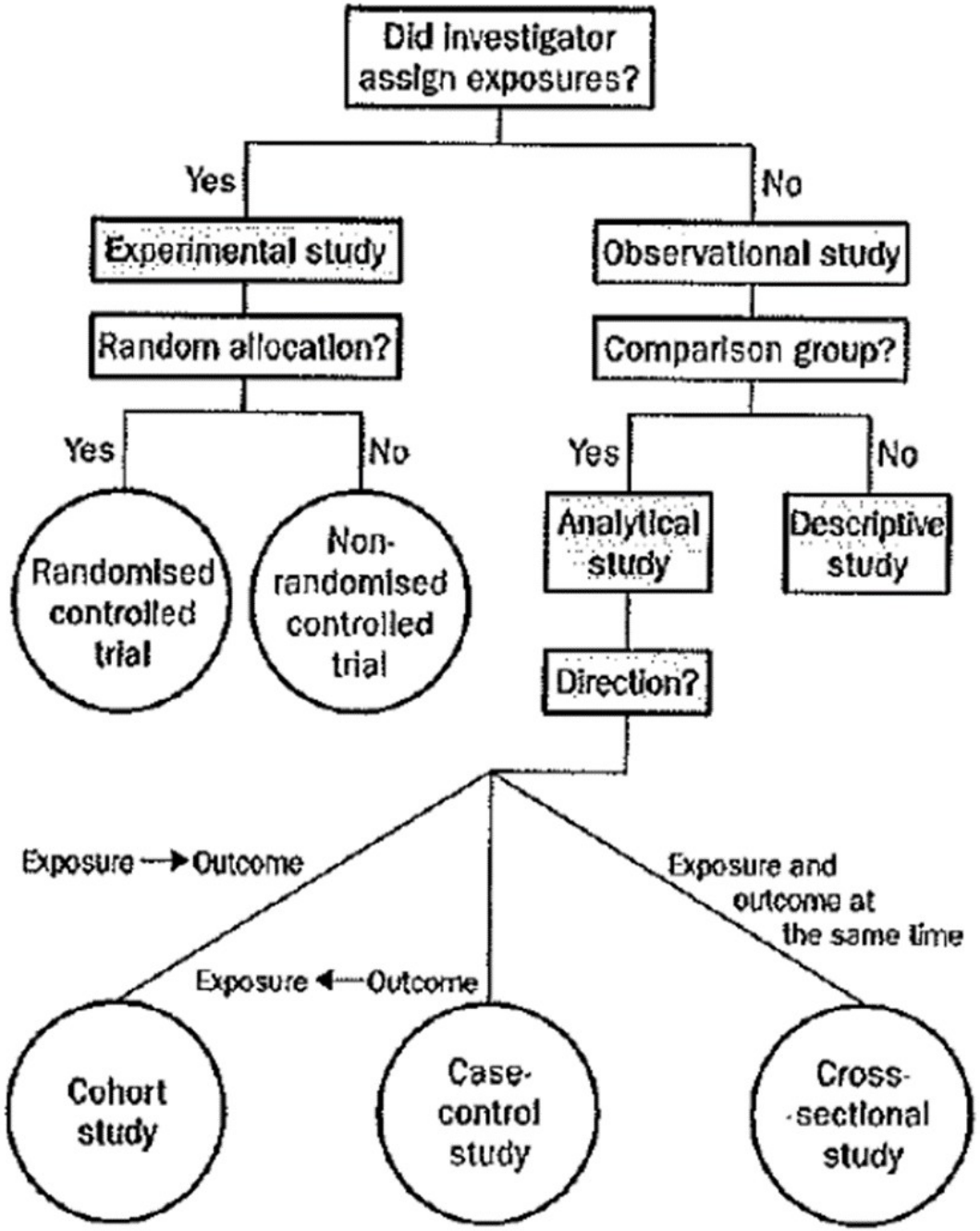
- ✓ Best type of study when objective is to establish efficacy of treatment or procedure i.e. provides strongest evidence for concluding causation
- ✓ Patients randomly assigned to different treatment arms ‘strongest’ design

x Expensive

x Long duration

x Need to be cautious about accepting practices without proper justification

x External validity may be limited


Did investigator assign exposures?
Yes
No
Experimental study
Observational study
Random allocation?
Comparison group?
Yes
No
Yes
No
Randomised controlled trial
Non-randomised controlled trial
Analytical study
Descriptive study
Direction?
Exposure → Outcome
Exposure ← Outcome
Exposure and outcome at the same time
Cohort study
Case-control study
Cross-sectional study

Systematic reviews

- Involve systematic literature searches to identify, appraise and synthesis all relevant research
- Used for evaluation of existing or new technologies and practices
- Attempts to reduce influence of bias
- Should be conducted no matter what the study design is
- Qualitative view

✓Justification based on scientific principle of replication

Meta-analyses

- Systematic, objective way to combine data from many studies (usually RCTs) and arrive at pooled estimate of treatment effectiveness and statistical significance
- Especially appropriate when studies that have been reported have small sample sizes
- Quantitative view
- Quality of the design of studies used needs to be assessed

✓ Increases sample size and allows for analyses that would not have been possible otherwise

x Publication bias

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Vaccines are not associated with autism: An evidence-based meta-analysis of case-control and cohort studies

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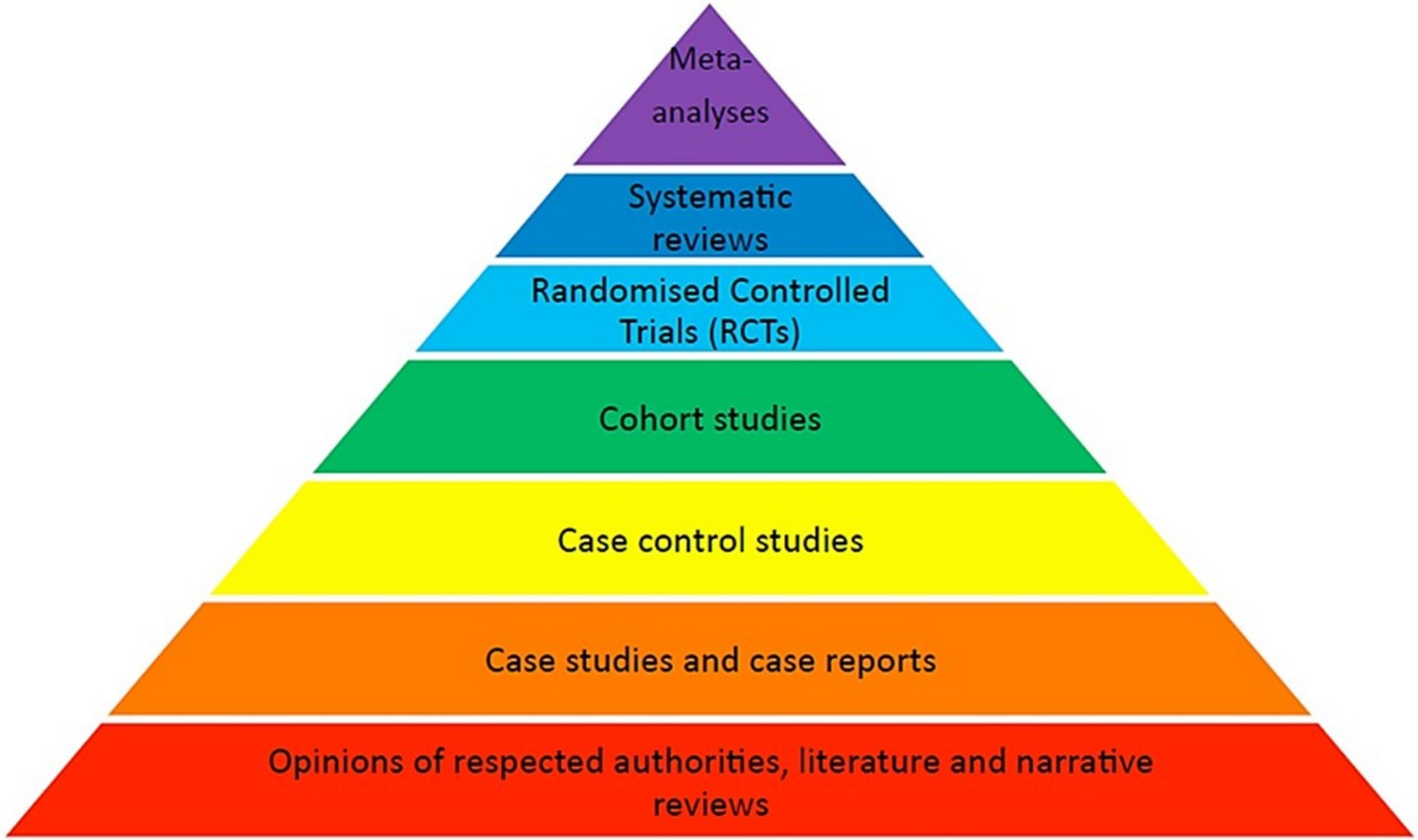
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Thimerosal
Mercury

ABSTRACT

There has been enormous debate regarding the possibility of a link between childhood vaccinations and the subsequent development of autism. This has in recent times become a major public health issue with vaccine preventable diseases increasing in the community due to the fear of a 'link' between vaccinations and autism. We performed a meta-analysis to summarise available evidence from case-control and cohort studies on this topic (MEDLINE, PubMed, EMBASE, Google Scholar up to April, 2014). Eligible studies assessed the relationship between vaccine administration and the subsequent development of autism or autism spectrum disorders (ASD). Two reviewers extracted data on study characteristics, methods, and outcomes. Disagreement was resolved by consensus with another author. Five cohort studies involving 1,256,407 children, and five case-control studies involving 9,920 children were included in this analysis. The cohort data revealed no relationship between vaccination and autism (OR: 0.99; 95% CI: 0.92 to 1.06) or ASD (OR: 0.91; 95% CI: 0.68 to 1.20), nor was there a relationship between autism and MMR (OR: 0.84; 95% CI: 0.70 to 1.01), or thimerosal (OR: 1.00; 95% CI: 0.77 to 1.31), or mercury (Hg) (OR: 1.00; 95% CI: 0.93 to 1.07). Similarly the case-control data found no evidence for increased risk of developing autism or ASD following MMR, Hg, or thimerosal exposure when grouped by condition (OR: 0.90, 95% CI: 0.83 to 0.98; $p = 0.02$) or grouped by exposure type (OR: 0.85, 95% CI: 0.76 to 0.95; $p = 0.01$). Findings of this meta-analysis suggest that vaccinations are not associated with the development of autism or autism spectrum disorder. Furthermore, the components of the vaccines (thimerosal or mercury) or multiple vaccines (MMR) are not associated with the development of autism or autism spectrum disorder.

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MMR vaccine controversy- linked MMR vaccine to autism spectrum disorders



The Hierarchy of Research[2]